

Lecture 4

*Lecturer: Nikhil Karamchandani**Scribe(s): Anway, Tamojeet*

1 Field

A field $\mathcal{F}$ is a set of elements with two operations defined on it - addition(+) and multiplication($\cdot$) that satisfy the following properties:

- $\mathcal{F}$ is an Abelian group under addition. The additive identity is denoted as 0.
- $\mathcal{F}$ is closed under multiplication. All elements of $\mathcal{F}$ other than 0 form an Abelian group under multiplication. The multiplicative identity is denoted by 1.
- Distributive law holds for both operations

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

$$\forall a, b, c \in \mathcal{F}$$

Subtraction can be defined as the addition of the additive inverse. If $(-b)$ denotes the additive inverse of b , then $a - b := a + (-b)$.

Similarly, all elements except 0 have a multiplicative inverse. Define $\frac{a}{b} := a \cdot (\frac{1}{b})$

1.1 Galois Fields

Consider the field $\mathcal{GF}(2) := \{0, 1\}$, with (+) and ($\cdot$) defined as mod-2 addition and multiplication respectively.

+	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

A similar field can be defined for any prime p , $\mathcal{GF}(p) := \{0, 1, 2, \dots, p-1\}$, with + and $\cdot$ defined as mod- p addition and multiplication respectively.

We can also create, for a prime p and positive integer q , the Galois field $\mathcal{GF}(p^q) := \{0, 1, 2, \dots, p^q - 1\}$

2 Vector Space over Finite Fields

If $\mathcal{F}$ is a finite field whose elements are scalars, then we say that $\mathbf{V}$ is a vector space over $\mathcal{F}$ if $\exists$ a vector addition operation (+) acting on pairs of elements from $\mathbf{V}$, and a scalar multiplication ($\cdot$) such that they satisfy the following properties:

- $\mathbf{V}$ should be an Abelian group under (+)

- $\forall a \in \mathcal{F}$ and $\forall v \in \mathbf{V}$, $a \cdot v \in \mathbf{V}$
- It satisfies distributivity $c \cdot (v_1 + v_2) = c \cdot v_1 + c \cdot v_2$ and $(c_1 + c_2) \cdot v = c_1 \cdot v + c_2 \cdot v$
- It satisfies associativity $(c_1 \cdot c_2) \cdot v = c_1 \cdot (c_2 \cdot v)$
- If 1 denotes the multiplicative identity in $\mathcal{F}$, then $1 \cdot v = v \forall v \in \mathbf{V}$
- If 0 denotes the additive identity in $\mathcal{F}$, then $0 \cdot v = \bar{0} \forall v \in \mathbf{V}$, where $\bar{0}$ is the additive identity in $\mathbf{V}$

Examples

1. A vector space over $\mathcal{GF}(2)$ can be collection of all n -tuples with elements drawn from the field, $\mathbf{V} = \{(a_0 a_1 \dots a_n) : a_i \in \mathcal{GF}(2)\}$ with $(+)$ defined as component-wise mod-2 addition and $(\cdot)$ defined as element-wise multiplication with the scalar
2. All polynomials with its coefficients belonging to $\mathcal{GF}(q)$ also form a vector space, with $(+)$ defined as polynomial addition and $(\cdot)$ defined as multiplication of the polynomial by the scalar
3. For a code of length n drawn using elements from some field $\mathcal{F}$, all possible codewords are all possible n -tuples of elements of $\mathcal{F}$ and hence form a vector space denoted as $\mathcal{F}^n$. The actual code may use only some specific codewords which form a subset of $\mathcal{F}^n$

2.1 Subspace

A subset $\mathbf{S} \subset \mathbf{V}$ is called a subspace if it satisfies the following properties:

- $\forall u, v \in \mathbf{S}, u + v \in \mathbf{S}$
- $\forall a \in \mathcal{F}, u \in \mathbf{S}, a \cdot u \in \mathbf{S}$

By replacing $a = 0$ in the second property, we see that the additive identity of $\mathbf{V}$, $\bar{0}$, must be present in any subspace.

A code whose codewords form a subspace over $\mathcal{F}^n$ is called a linear code. For example, let $\mathbf{V}$ be a space of 5-tuples over $\mathcal{GF}(2)$. Then the set $\mathbf{S} = \{00000, 00111, 11010, 11101\}$ forms a linear code. Here addition is simply the bitwise XOR operation.

If we choose some k arbitrary vectors v_1 to v_k from $\mathbf{V}$, then all their possible linear combinations $\sum_{i=1}^k a_i v_i \forall a_i \in \mathcal{F}$ forms a subspace of $\mathbf{V}$.

2.2 Linear Dependence, Inner Product and Orthogonality

For a set of k vectors v_1 to v_k from $\mathbf{V}$, if $\exists a_1, a_2 \dots a_k \in \mathcal{F}$, not all of them equal to 0, such that $\sum_{i=1}^k a_i v_i = 0$, then those vectors v_1 to v_k are said to be linearly dependent. If no such a_1 to a_n exist, then those vectors are linearly independent.

Let $u = u_0 u_1 \dots u_{n-1}$ and $v = v_0 v_1 \dots v_{n-1}$ be two vectors in $\mathcal{GF}(2)^n$. Then their inner product is denoted as $\langle u, v \rangle$ or $u \cdot v$ and defined as $u \cdot v = \sum_{i=0}^{n-1} u_i v_i$.

If $u \cdot v = 0$, then they are said to be orthogonal. For eg. Consider a vector $v = (1,1)$ over $\mathcal{GF}(2)$, vectors orthogonal to v are $(0,0)$ as well as $(1,1)$.

Claim: $\mathbf{V}$ is a vector space, $\mathbf{W}$ is a subspace of $\mathbf{V}$, then the set of vectors orthogonal to $\mathbf{W}$ also form a **subspace**. We define this as the **Orthogonal Complement** or the **Nullspace** of $\mathbf{W}$, denoted as $\mathbf{W}^\perp$.

Properties:

- $(\mathbf{W}^\perp)^\perp = \mathbf{W}$.
- if $\dim(\mathbf{W}) = k$, then $\dim(\mathbf{W}^\perp) = n - k$.

For eg. Consider a vector space over $\mathcal{GF}(2)$ consisting of all vectors of length 3. Now, consider $\mathbf{W} = \{000, 101, 001, 100\}$. So, $\mathbf{W}^\perp$ will have $\{000, 010\}$. We can confirm that $\dim(\mathbf{W}) = 2$ and $\dim(\mathbf{W}^\perp) = 3 - 2 = 1$.

2.3 Matrices

Consider a $k \times n$ matrix $\mathbf{G}$ over $\mathcal{GF}(q)$:

$$\mathbf{G} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \dots & a_{kn} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_k \end{bmatrix}$$

$\mathbf{G}$ has k rows $a_i \in V$. We define the Rowspace of $\mathbf{G}$ as the set of all linear combinations of the rows of $\mathbf{G}$. So, we can say that any row transformation on $\mathbf{G}$ won't change its Rowspace. So, let us consider a matrix with rowspace $(\mathbf{G}) = \mathbf{W}$ and all rows of $\mathbf{G}$ to be linearly independent. Thus, $\dim(\mathbf{W}) = k$. Now, consider that $\mathbf{W}^\perp$ has basis $\{h_1, h_2, h_3, \dots, h_{n-k}\}$. Construct

$$\mathbf{H} = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_{n-k} \end{bmatrix}$$

Here, $\text{Rowspace}(\mathbf{H}) = \mathbf{W}^\perp$. Hence, for any $g_i \in \mathbf{G}, h_j \in \mathbf{H}$ we have $g_i h_j = 0$.

For a Code, we can construct a generator matrix $\mathbf{G}$ and parity checker matrix $\mathbf{H}$.

3 Block Linear Code

Consider a Field $\mathcal{F} = \mathcal{GF}(q)$. An (n, M, d) code $\mathcal{C}$ over $\mathcal{F}$ is Linear if $\mathcal{C}$ is a subspace of $\mathcal{F}^n$ i.e. $\forall c_1, c_2 \in \mathcal{C}$ and $a_1, a_2 \in \mathcal{F}$, $a_1 c_1 + a_2 c_2 \in \mathcal{C}$.

Let us say $\dim(\mathcal{C}) = k$. Then $|M| = q^k$. So the rate of this code becomes $\frac{\log_q |M|}{n} = \frac{k}{n}$. Let us see this for some specific codes:

- Simple parity checker (3,4,2) code. So, $\mathcal{C} = \{(000), (011), (101), (110)\}$. Since $\mathcal{C}$ is subspace of $\{0,1\}^3$, its a valid linear code. $\dim(\mathcal{C}) = 2$. So, a basis can be $\{(011), (101)\}$, and rate of the code is $2/3$.

- Repetition code (3,2,3). So, $C = \{ (000), (111) \}$. It is a valid linear code. $\dim(C) = 1$. A basis can be $\{ (111) \}$, and rate = $1/3$.

Let us try to create a generating matrix for these. We can construct a generating matrix using the basis as rows of the matrix. Hence

$$G_1 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

and G_2 can be $[1 \ 1 \ 1]$.

3.1 Minimum Distance

We can define minimum distance for C in a different way:

$$d = \min_{\substack{c \in C \\ c \neq 0}} w(c)$$

$w(c)$ is called the Hamming weight of c and is the number of non zero entries in c .

Proof:

$$d = \min_{\substack{c_1, c_2 \in C \\ c_1 \neq c_2}} d(c_1, c_2) = \min_{\substack{c_1, c_2 \in C \\ c_1 \neq c_2}} w(c_1 - c_2) = \min_{\substack{c \in C \\ c \neq 0}} w(c)$$